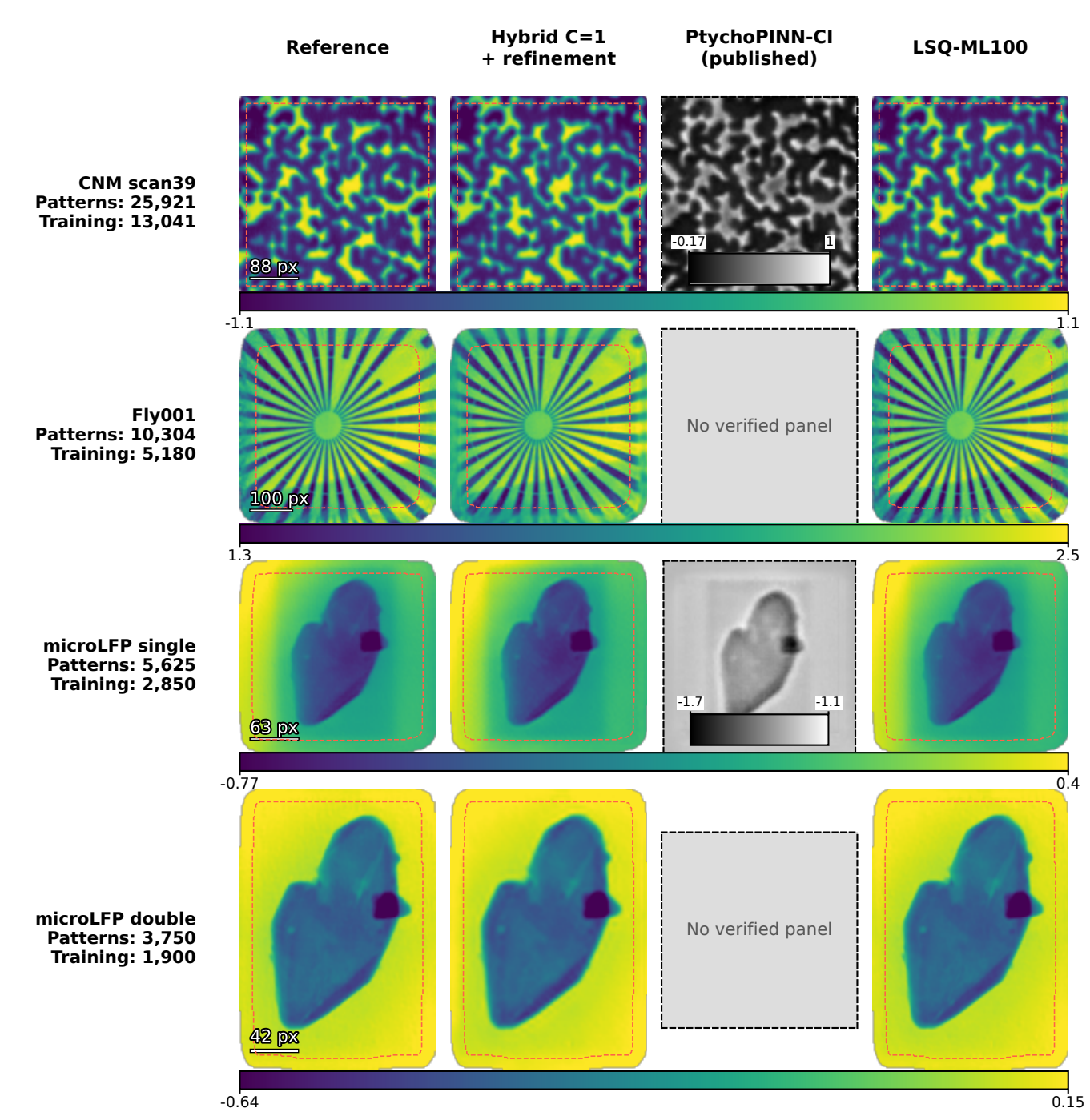
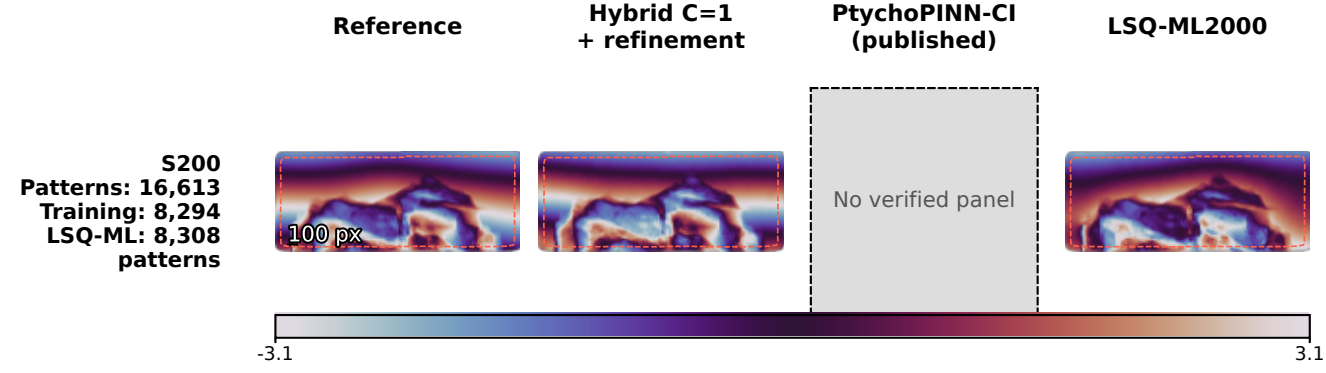


Phase is shown in radians; amplitude shows relative object amplitude. Pattern counts refer to the displayed acquisition and distinct training patterns. High half selects checkpoints (spatial validation, not an independent test; no guard band). Reference is an experimental estimate; dashed contour marks scored interior. Prior fragments: Vong et al., Contrast-invariant deep ptychography neural networks, arXiv:2608.02869, synthetic-only whole-field crops, cropped and rescaled (CC BY 4.0). Their original scales are shown in-cell; the shared computed-method bar excludes prior fragments. Whole-acquisition images include the training region. Paired groups assign C4 only where it has lower phase MAE after refinement and higher phase SSIM; other datasets use C1. This is the C1-selected group. S200 LSQ-ML2000: historical 8,308-pattern subset versus 16,613 neural patterns; iteration-budget limited, not a matched timing result.



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